

RESEARCH INTERESTS

Atom-Light Interactions, Neutral-Atom Arrays, Quantum Simulation and Computing, Quantum Metrology

EDUCATION

Kyungpook National University (KNU)

Bachelor of Science in Physics

Overall GPA: 4.11/4.3 | Major GPA (71 credits): 4.19/4.3

Daegu, Republic of Korea

Mar. 2020 – Feb. 2024

RESEARCH EXPERIENCE

Agency for Defense Development (ADD)

Research Officer for National Defense (Military service)

Daejeon, Republic of Korea

Jun. 2024 – Present

Research in experimental atomic physics and quantum sensing (PIs: Dr. Sangkyung Lee and Dr. Sin Hyuk Yim)

- Developed an automated absorption spectroscopy system with spectral modeling to characterize atomic vapor cells and contributed to their fabrication [1]
- Built a zero-field optically pumped magnetometry platform to evaluate component-induced magnetic interference and provide design guidance for portable atomic sensors [4]
- Deployed a drone-mounted atomic magnetometer for magnetic anomaly detection and established a quantitative framework for interpreting the acquired data [2]

Korea Advanced Institute of Science and Technology (KAIST)

Research Intern (MiliTECH Challenge)

Daejeon, Republic of Korea

- **Quantum and Nanobio Photonics Lab** (Advisor: Prof. Yong-Hoon Cho) Apr. 2023 – Dec. 2023
Project: Semiconductor quantum dot-optical fiber coupling for quantum cryptographic communication
- **Aerospace Systems and Control Lab** (Advisor: Prof. Hyo-choong Bang) Apr. 2022 – Dec. 2022
Project: Design of air-land quadrotor using hemisphere shell passive wheel

Kyungpook National University (KNU)

Undergraduate Researcher

Daegu, Republic of Korea

- **Novel Applied Nano Optics Lab** (Advisor: Prof. Junyeob Yeo) Jun. 2021 – Jun. 2022
Fabricated a flexible photoelectrochemical cell using laser processing
- **High Energy Physics Lab (Moon Lab)** (Advisor: Prof. Chang-Sung Moon) Jun. 2020 – Feb. 2021
Developed momentum reconstruction algorithms for silicon detector simulations

PUBLICATIONS

[†] Corresponding author

- [4] **H. Kim**, S. Lee[†], S. H. Yim, T. Jeong, and Y. Lim, “Magnetic characterization of electronic components for portable atomic sensors using a zero-field optically pumped magnetometry platform,” (submitted); *arXiv:2609.07270* (2026). [[paper](#)]
- [3] S. H. Hong, Y. Lim[†], S. H. Yim, J. B. Nam, S. Lee, T. Jeong, and **H. Kim**, “Wire-sealed atomic gas cells for noble-gas nuclear magnetic resonance applications,” (under review)
- [2] **H. Kim**, S. H. Yim[†], S. Lee, E.-S. Bang, S. R. Yeo, Y. J. Kim, T. Jeong, and Y. Lim, “Magnetic Anomaly Detection of an Undersea Tunnel Using a Drone-Mounted All-Optical Pulsed Atomic Magnetometer,” *IEEE Sensors Journal* Early Access (2026). [[paper](#)]
- [1] **H. Kim**, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim[†], “Aging test of atomic vapor cell with Al₂O₃ wall coating on cubic glass,” *Applied Optics* **64**, 7932–7937 (2025). [[paper](#)]

SELECTED PRESENTATIONS

[†] Corresponding author | Full list and abstracts [[link](#)]

H. Kim, S. Lee[†], Y. Lim, S. H. Hong, T. Jeong, and S. H. Yim, “Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer,” *APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting* (2026), poster

H. Kim[†], “Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells,” *KISTI-SNU Joint Workshop* (2026), invited talk

H. Kim, T. Jeong, S. Lee, and S. H. Yim[†], “Lifetime extension of rubidium vapor cells by Al₂O₃ coating,” *APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting* (2025), poster

SOFTWARE COPYRIGHTS

[†] Lead inventor

H. Kim[†], S. Lee, T. Jeong, and S. H. Yim, “Buffer-Gas Pressure Estimation Software for Quantum Magnetometer Vapor Cells,” *Korea Copyright Commission* C-2024-047156 (2024)

H. Kim[†], S. H. Yim, T. Jeong, and S. Lee, “Rubidium Vapor Cell Lifetime Assessment Software,” *Korea Copyright Commission* C-2024-054414 (2024)

HONORS AND AWARDS

KFAS Study Abroad Workshop, *Korea Foundation for Advanced Studies* Jul. 2026

Physics Alumni Association Award (Outstanding Graduate), *KNU Physics Alumni Association* Feb. 2024

2nd Prize at the 2023 MiliTECH Challenge, *KAIST* Dec. 2023

2nd Prize at the 2022 MiliTECH Challenge, *KAIST* Dec. 2022

3rd Prize at the Academic Conference, *KNU College of Natural Sciences* Dec. 2021

Dean’s List, *KNU College of Natural Sciences* Nov. 2021, Apr. 2022, Apr. 2023

SCHOLARSHIPS

National Scholarship for Science and Engineering, *Korea Student Aid Foundation* Mar. 2022 – Feb. 2024

- Full tuition and \$8,000 in stipends

Hyoseok Scholarship, *KNU Alumni Association* Apr. 2022

- Selected as the College of Natural Sciences representative; awarded \$1,600

Academic Excellence Scholarship, *KNU Department of Physics* Aug. 2020, Feb. 2021, Aug. 2021

TEACHING EXPERIENCE

Teaching Assistant, *KNU Undergraduate Tutoring Program* Mar. 2023 – Dec. 2023

- Gave weekly two-hour lectures for Electromagnetism II (Spring) and Classical Mechanics II (Fall)

Tutor, *KNU International Student Tutoring Program* Feb. 2022 – May 2022

- Provided weekly two-hour campus-life guidance and calculus tutoring; selected as an Outstanding Tutor

SKILLS

- Programming: Python, Wolfram Language
- Software: LabVIEW, SolidWorks, COMSOL Multiphysics, Origin